DEMAND AND SUPPLY ANALYSIS

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Outline

- Demand and supply
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- Behaviour of Buyers (Demand)
- Individual and Market Demand
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- Changes in Demand and Quantity Demand
- Behaviour of Sellers (Supply)
- Individual and Market Supply
- Non-Price Determinants of Supply
- Market Equilibrium
- Mathematical Approach to demand and Supply Analysis

Demand and Supply Analysis

- In this lecture,
  - we examine how the forces of demand and supply serve to establish “normal” prices in markets
  - We show how the equilibrium of demand and supply in the market maximizes the total benefits received by buyers and sellers
  - We examine how changes in the demand and supply affects quantity demanded (supplied) and prices of goods and services

Markets

- *Market* - any set of arrangements that allows buyers and sellers to communicate and arrange exchange of goods and services.

- There exist markets for
  - Commodities (commodity/goods and services market)
  - Goods and Services (Goods and services market)
  - Houses (the housing market)
  - Stocks (the stock market)
  - Money (the money market)
  - Currencies (the forex market) etc.

Markets

- Where no government intervention occurs in the exchange of these goods and services, then we can have a *free market*.
- In the *free market*, no one issues commands or provides direction, but instead the market acts in the following ways:
  - It allocates resources to the production of different goods and services
  - It determines what resource mix to employ
  - It distributes the goods and services among households
- It is the *invisible hand* (according to Adam Smith) of self-interest that acts as the guiding force in this market

Markets

- Markets reduce the
  - cost of bringing buyers and sellers together
    - the transaction cost of exchange
  - cost of finding out what is available for sale, its price, its quantity and how it compares with other items for sale
    - » the cost of time and information required for exchange

Analyzing the Behaviour of Buyers (Demand)

- What determines how much goods people are willing to buy at any particular time?
- The most important factor determining how much goods people buy is the ***price*** of the good.
- Therefore, the more expensive it is for one to buy the good, the less he will be prepared to buy and vice versa *ceteris paribus*

Analyzing the Behaviour of Buyers (Demand)

- This behaviour represents what we call the ***law of demand*** which states that, *ceteris paribus* (i.e., everything else remaining constant)
  - at higher prices, a lower quantity will be demanded
  - at lower prices, a higher quantity will be demanded

- This is the case because when the price of the good rises two things happens
  - *relative prices* and *real income* of potential buyers change
- Buyers take both into account when they react to the price change

Analyzing the Behaviour of Buyers (Demand)

- The change in the relative price means that the good is now more expensive when compared with other similar goods.
  - buyers react to this by choosing to purchase less of it.
  - This condition is referred to as ***substitution effect*** of a price change.

- The increase in the price of the good also reduces buyers real income
  - This implies they can now only buy less than before with their nominal income

Analyzing the Behaviour of Buyers (Demand)

- The way buyers react to a change in their real income depends on the nature of the goods.
  - For ***normal goods***, their demand increases when income increases and falls when income decreases
  - For ***inferior goods***, their demand falls when income increases and increases when income decreases.
- These conditions are referred to as the ***income effects*** of a price change.
- Thus, the law of demand can be explained using the ***income*** and ***substitution effects***.
- The relationship between price and quantity demanded of a good or service is shown next graphically;

Analyzing the Behaviour of Buyers (Demand)

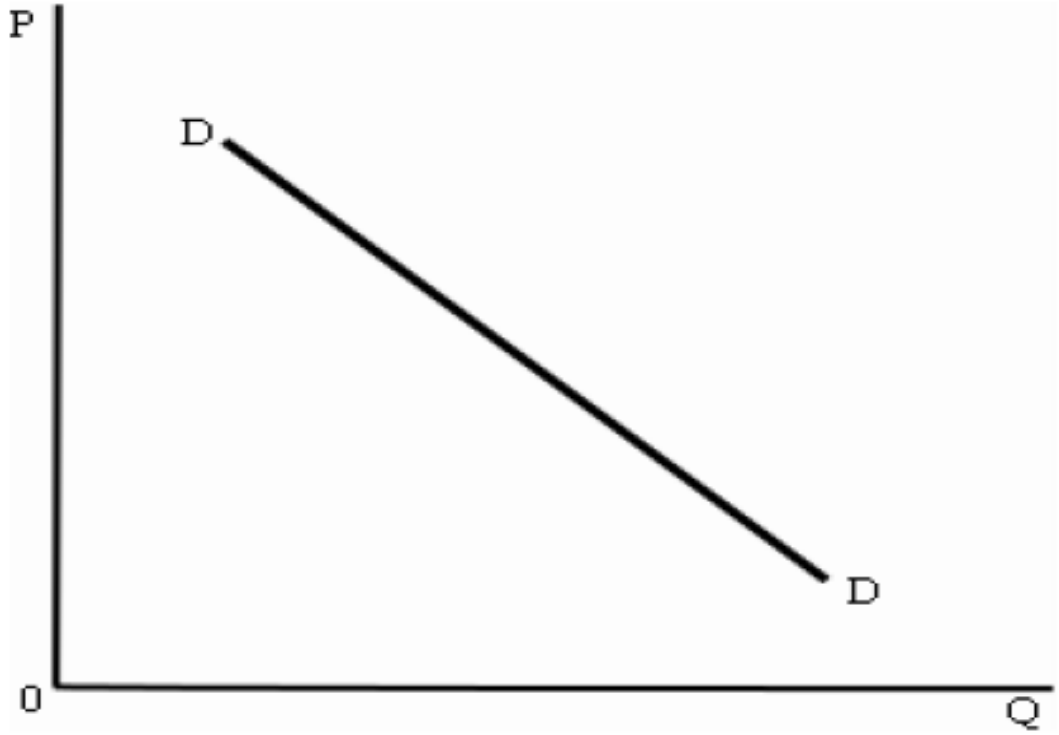


Figure 2.1: The demand for a good (demand curve)

Analyzing the Behaviour of Buyers (Demand)

- Demand can therefore be defined as
  - the quantity of a good/service that buyers are able and willing to purchase in a market at a given price during a specified period, *ceteris paribus*.
- Here, price and quantity demanded move in opposite directions.
- Individual demand talks about the demand of an individual entity.
  - Its curve is like the one we saw earlier.
- The ***market demand*** of a good or service is simply the sum of the demands of all buyers in a market at a given price during a specified period, *ceteris paribus*.

Individual and Market Demand

- The market demand traces the relationship between the price of a good and the quantity that all buyers in the market are willing and able to buy. This can be shown in the next diagram
- The demand curve in panel (c) is obtained by summing up horizontally the individual demand curves in panel (a) and (b).

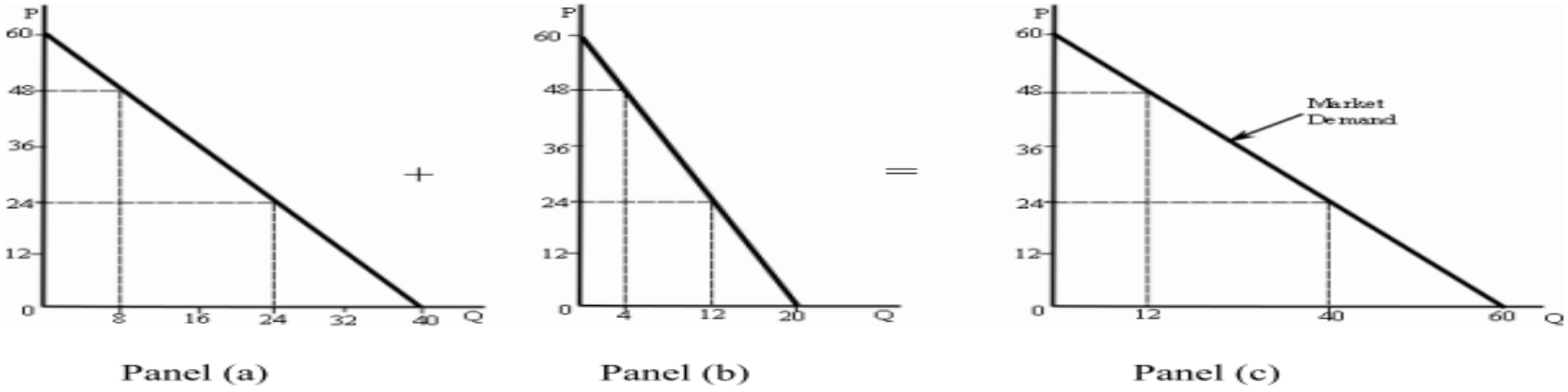


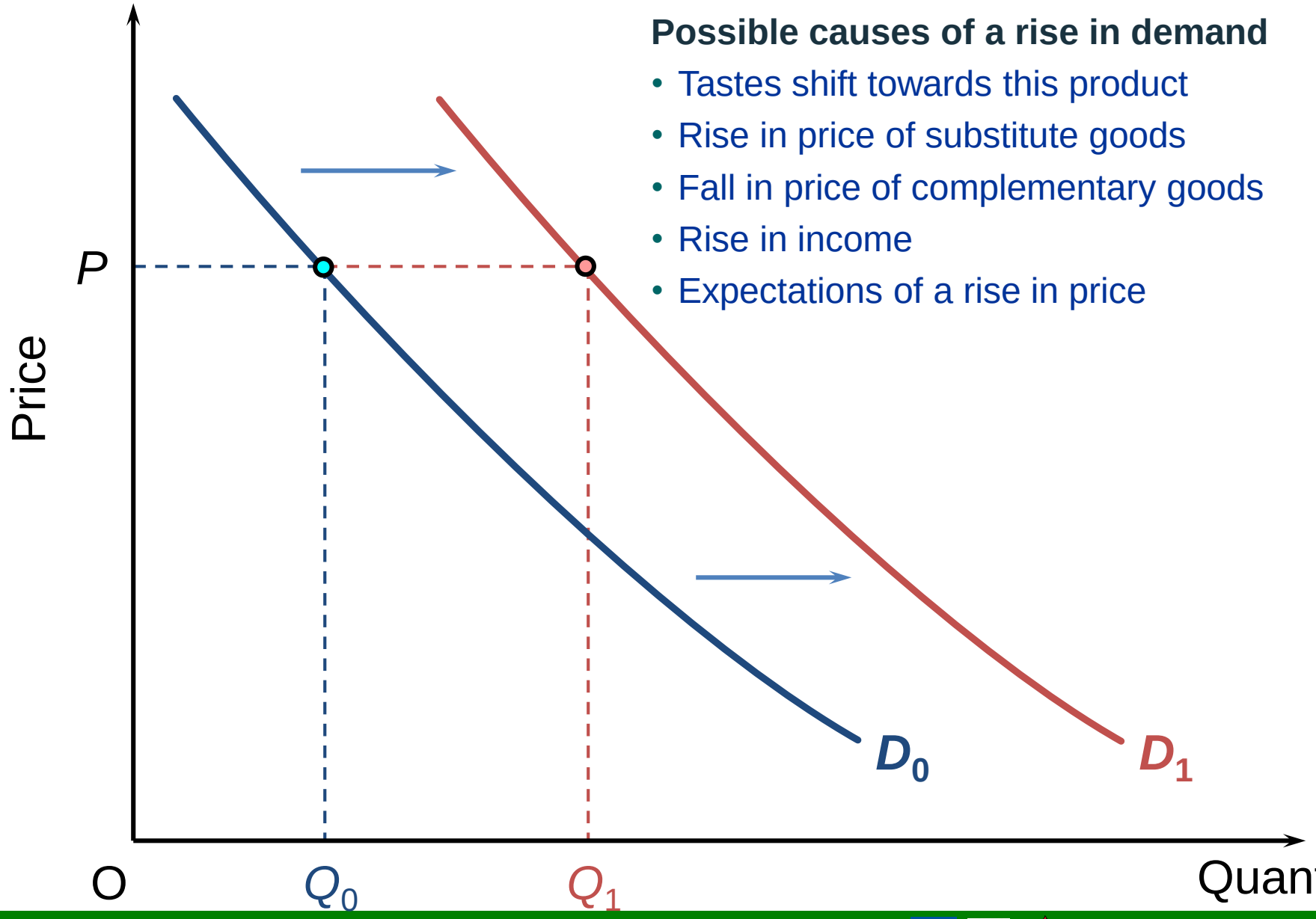
Figure 2.2: The market demand for a good

Non-Price Determinants of Demand

- Apart from price, the following are some of the non-price determinants of demand;
  - Income
  - Distribution of income
  - Taste and preferences
  - The *number* and *price* of substitute goods and complements
  - Weather condition
  - Number of potential buyers (Population)
  - Government Policy(Taxes and Subsidies)

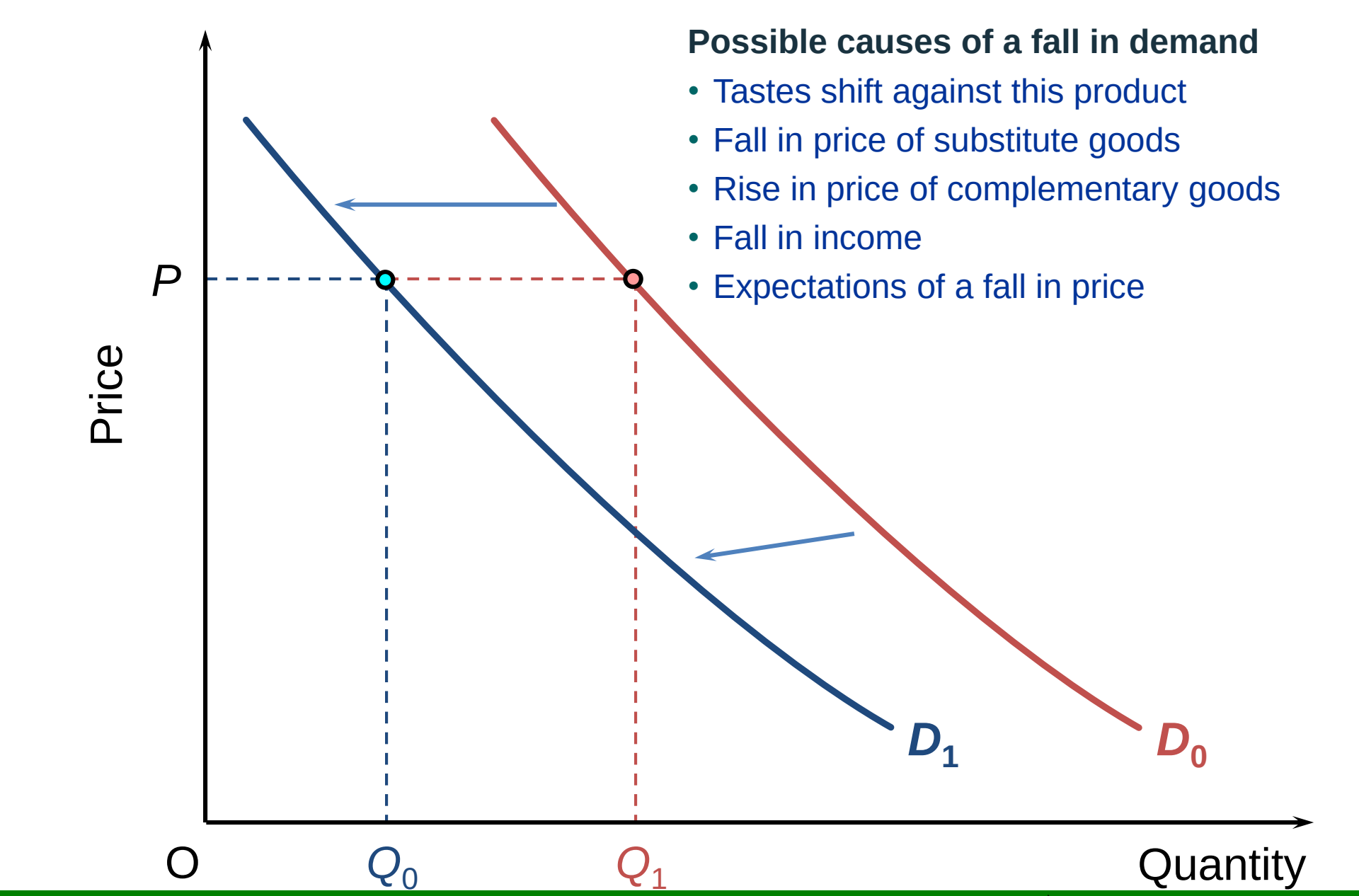
Non-Price Determinants of Demand

- Whenever there is a change in any of these non-price determinants of demand,
  - the whole demand curve shifts
- Assuming D_0 is the original demand curve.
  - then, if a change in one of the non-price determinants causes an increase in demand – say, income rises
    - the whole demand curve shift to the right (i.e. from D_0 to D_1)
- On the other hand,
  - if a change in a determinant other than price causes demand to fall – say, income falls,
    - the whole demand curve shift to the left (i.e. from D_0 to D_2).



Possible causes of a rise in demand

- Tastes shift towards this product
- Rise in price of substitute goods
- Fall in price of complementary goods
- Rise in income
- Expectations of a rise in price


Possible causes of a fall in demand
• Tastes shift against this product
• Fall in price of substitute goods
• Rise in price of complementary goods
• Fall in income
• Expectations of a fall in price
P
Price
O
Q_0
Q_1
Quantity
D_1
D_0

Change in Demand vs. Change in Quantity Demanded

- ***Changes in demand*** result from shifts in the entire demand curve
  - such that the quantity demanded at each and every price changes
  - these are caused by the non price determinants of demand.

- ***Changes in the quantity demanded*** result only from a change in price and they are represented by a movement along the same demand curves.

Analyzing the Behavior of Sellers (Supply)

- *Supply* is defined as the quantity of a good or service that sellers are willing and able to sell in a market at a given price during a specified period, *ceteris paribus*.
- This positive relationship is shown graphically as the supply curve in the next diagram.

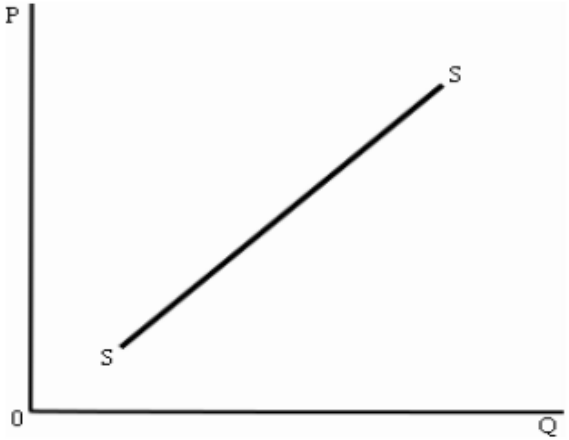


Figure 2.4: The supply of a good (supply curve)

Individual and Market Supply

- *Individual supply* simply shows the amount of a good or service in which a seller is able to supply at a given price during a specified period, *ceteris paribus*.
- *Market supply* of a good or service is simply the horizontal sum of the supply of all sellers in a market at a given price during a specified period, *ceteris paribus*.
  - It traces the relationship between the price of a good or service and the quantity that all sellers in the market are willing and able to sell.

Individual and Market Supply

- The *supply curve* reflects the *law of supply* which states that
  - the quantity offered for sale is positively related to the price;
  - at higher prices, larger amounts are supplied and at lower prices, smaller amounts are supplied, *ceteris paribus*.

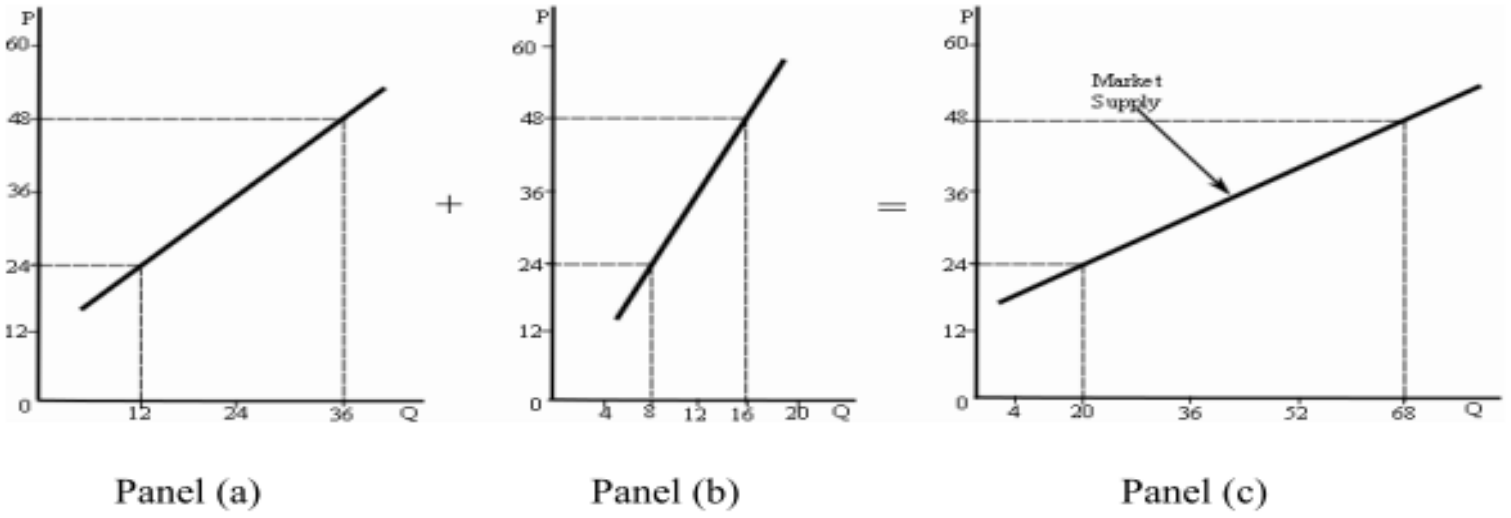


Figure 2.5: The market supply of a good

Why is the supply curve upward-slopping or the positive relationship?

- Sellers offer more goods for sale when prices are higher for three important reasons.
  - the higher the price of a good or service, the more profitable it becomes to produce and supply
  - through the theory of increasing costs
    - Firms must spend more as they supply more at some point in time
- Over time, if the price of the good or service remains high, then new producers will be encouraged to set up in producing the same good or service and total market supply will thus rise.

Non-Price Determinants of Supply

- Like demand, supply is not only determined by price.
- The non-price determinants are
  - *costs of production*
  - *profitability of alternative products*
  - *the number of suppliers*
  - *unpredictable events*
  - *The level of technology*
  - *Price expectation*
  - *Government policy(taxes and subsidies)*

Non-Price Determinants of Supply

- Whenever there is a change in any of these non-price determinants of supply, the whole supply curve shifts(Changes in Supply)
- Whenever there is a change in only price, there is a movement along the same supply curve (Changes in Supply)
- As shown in the next figure;

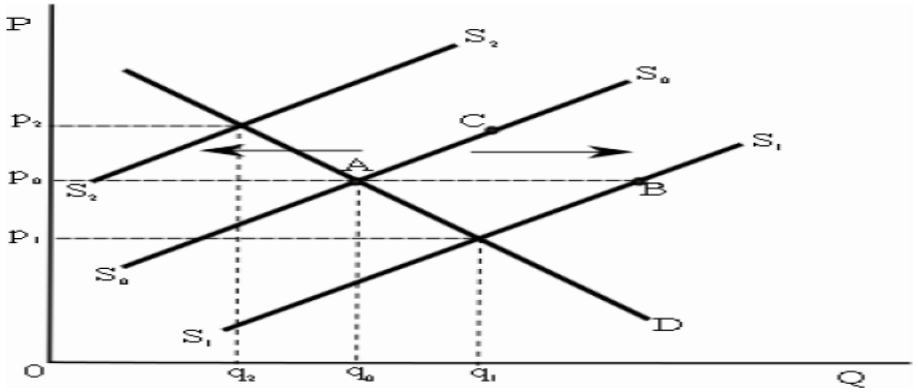


Figure 2.6: Shifts in market supply for a good

Market Equilibrium

- This is the *market clearing* or *equilibrium price* and it occurs at the point of intersection of the supply and demand curves.
- At this equilibrium price, the market clears leaving no shortage or surplus.
- Thus at the equilibrium price, the quantity demanded is the same as the quantity supplied.

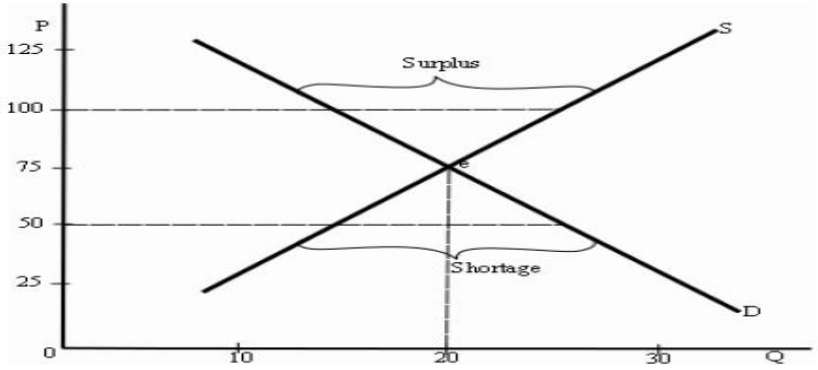


Figure 2.7: The Determination of Market Equilibrium

Market Equilibrium

- At prices below the equilibrium level, the quantity demanded by buyers is more than the quantity of goods or service that suppliers are willing and able to sell.
- This results in a temporary shortage
- At prices above the equilibrium level, the quantity of the good or service that sellers wish to supply is greater than the quantity buyers wish to purchase.
- This results in a temporary surplus.

A Mathematical Approach to Equilibrium Analysis

- The demand and supply curves can be represented by the following equations;

$$Q_D = a - bP$$

$$Q_S = c + dP$$

$$c + dP = a - bP$$

$$Q_e = \frac{ad + ab - ba + bc}{d + b} = \frac{ad + bc}{d + b}$$

$$P = \frac{a - c}{d + b}$$

Market Equilibrium and Efficient Allocation of Resources

- The next figure can help us understand the idea of; allocative efficiency of market equilibrium in terms of
  - *consumer surplus*
  - *producer surplus*

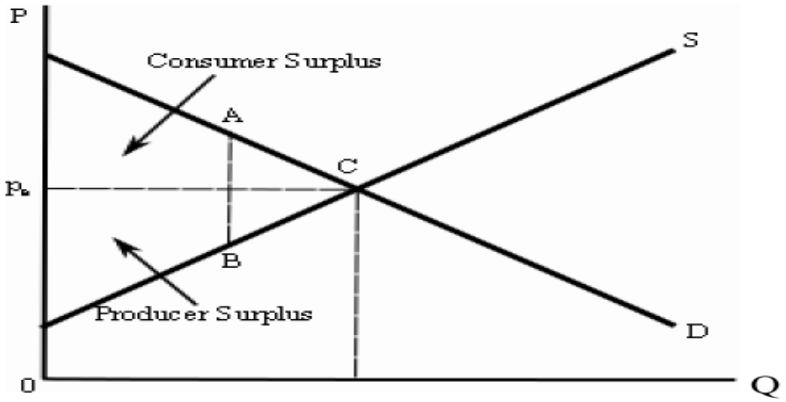


Figure 2.8: Consumer and Producer Surplus in the Market Equilibrium

Market Equilibrium and Efficient Allocation of Resources

- Consumer surplus represent the excess of what buyers/consumers are willing to pay over what they pay.
- It can be computed by finding the area below the demand curve and above the price.
- *Producer surplus* represent the amount sellers receive for their goods minus the minimum amount they are willing to accept.
- We can obtain this by finding the area below the price and above the supply curve.

Market Equilibrium and Efficient Allocation of Resources

- A measure of the sum of *consumer surplus* and *producer surplus* is known as *total surplus* in a market.
- If the quantity is below the equilibrium level of output (say at the level corresponding to line *AB*), the total surplus is not at a maximum, and its size can be increased by the area ABC.

The End



Next Time: What Money Is About?